Sentiment Analysis of Code-Mixed Singlish (Sinhala-English) Reviews with Emoji Fusion NLP Approach

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***Abstract*—Sentiment analysis of code-mixed Sinhala-English text — commonly termed "Singlish" — has recently begun to attract research attention, with annotated datasets and fine-tuned multilingual transformers (e.g., XLM-R) now reported for domains such as banking feedback and e-commerce product reviews [Aththanayaka and Naleer, 2020, Rizvi et al., 2025, Galwatta et al., 2026, Dhananjaya et al., 2022]. However, this emerging body of work consistently treats emoji as noise to be stripped during preprocessing rather than as a signal to be modeled, and the existing Daraz.lk-based dataset targets aspect-based extraction rather than emoji-fused overall sentiment. Separately, emoji-aware sentiment fusion has been explored for other code-mixed language pairs, such as English-Hindi [Singh et al., 2024], but not for Sinhala-English. This research targets that specific, verified gap: it proposes to (i) construct an annotated corpus of Singlish reviews that retains emoji, drawing on Daraz.lk product reviews and currently unstudied Sri Lankan mobile app store reviews; (ii) fine-tune multilingual transformer models on this corpus, building on established Sinhala transformer benchmarking [Dhananjaya et al., 2022]; and (iii) design and evaluate explicit emoji-text fusion strategies, benchmarked against text-only baselines. The expected contributions are a new emoji-preserving annotated dataset spanning e-commerce and app-store reviews, and the first controlled emoji-fusion benchmark for Sinhala-English code-mixed sentiment analysis.**

Recent work has established that Sinhala-English code-mixed sentiment analysis is feasible: annotated corpora exist [Aththanayaka and Naleer, 2020], XLM-R has been benchmarked as the strongest available multilingual encoder for Sinhala [Dhananjaya et al., 2022], and fine-tuned XLM-R models now report 88–92% accuracy on Sinhala and code-mixed banking feedback [Rizvi et al., 2025]. A concurrent aspect-based sentiment dataset built from Daraz.lk product reviews, SalAngaBhava, further closes the e-commerce review gap [Galwatta et al., 2026]. What remains unsolved is narrower than originally assumed: (1) every one of these Sinhala-English studies removes or ignores emoji during preprocessing, so no existing work measures whether emoji carry sentiment signal beyond what the code-mixed text already encodes, or how best to fuse the two; (2) emoji-text fusion architectures have been demonstrated for other code-mixed pairs, e.g. English-Hindi [Singh et al., 2024], but this design has not been adapted or evaluated for Sinhala-English; and (3) SalAngaBhava's Daraz.lk coverage is aspect-based rather than emoji-fused overall sentiment, and Sri Lankan mobile app store reviews — an emoji-dense source in practice — have no published annotated sentiment data at all. (Ride-hailing platforms such as PickMe were considered as an additional source but do not expose a public review/comment feature, so they are excluded from the data collection plan.) This research is scoped to these three specific gaps rather than to code-mixed Sinhala sentiment analysis in general.

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